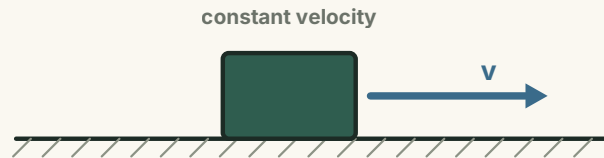


Newton's First Law

Motion does not need a force to continue. A change in motion does.

ESSENTIAL QUESTION

If a moving object naturally keeps moving, why do everyday objects slow down and stop?



● LESSON MAP

Replace an everyday illusion with a precise rule.

01

QUESTION

Does force sustain motion,
or does friction hide the
truth?

02

IDEALISE

Reduce resistance and
extend Galileo's pattern.

03

STATE

Connect zero net force to
constant velocity.

04

APPLY

Use inertia to predict what
happens next.

A SCIENTIFIC HABIT

When observation and explanation disagree, find the **hidden influence** before changing the rule.

OBSERVE

A coasting scooter slows down — but why?

The same observation supports two explanations. Only one survives a better test.

FIRST GUESS

"Motion runs out without a push." This treats force as something that must continually maintain velocity.

BETTER EXPLANATION

Resistance changes the motion. Rolling resistance and air drag give a backward net force, so the scooter slows.

QUESTION

Friction can disguise an object's natural motion.

Galileo challenged the old conclusion by changing the conditions of the test.

01 · Everyday observation

Objects on rough surfaces slow down quickly.

02 · Change one influence

Make the surface smoother, so friction is smaller.

03 · Follow the pattern

With less friction, the object travels farther.

04 · Idealise

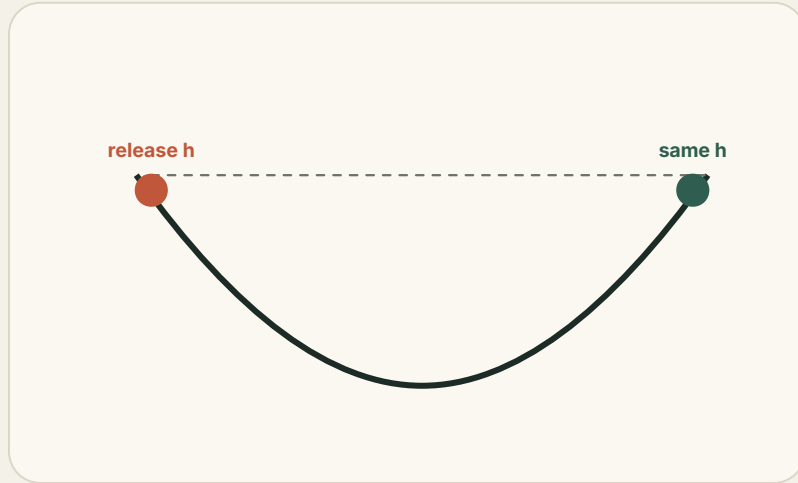
Ask what happens in the limit of zero resistance.

THE SHIFT A force is needed to overcome resistance — not to sustain motion.

IDEALISE

On a frictionless track, the ball regains its height.

The upward slope changes the direction of motion, but the target height stays the same.

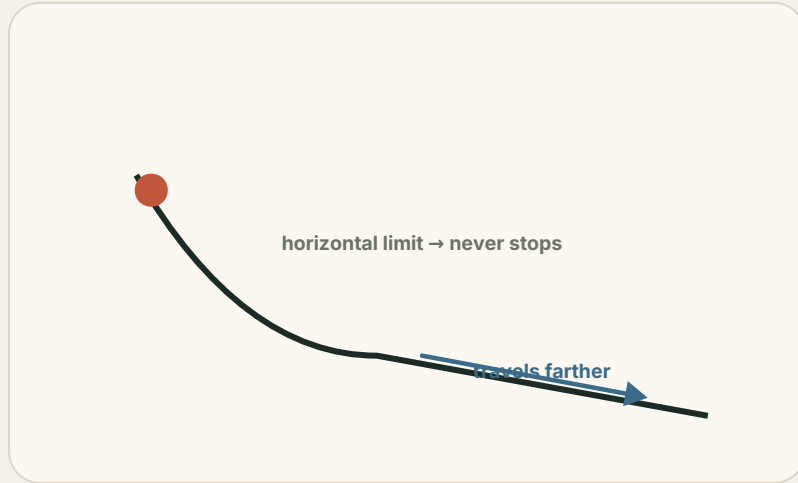


With no friction, mechanical energy is not dissipated — so the ball climbs back to where it began.

EXTRAPOLATE

Flatten the second slope, and the ball travels farther.

In the horizontal limit it can never regain the height — so it never stops.



The ideal experiment combines real evidence with logical extrapolation.

THE CONCLUSION

With no resistance, constant velocity continues forever.

STATE

Zero net external force means constant velocity.

In an inertial frame, rest is simply constant velocity equal to zero.

$$\sum \vec{F} = 0 \Leftrightarrow \vec{v} = \text{constant}$$

AT REST

Position stays unchanged.

MOVING

Speed and direction stay unchanged.

INTERPRET

Force changes velocity — not merely position.

A net force is required for speeding up, slowing down, or turning.

01 · Speed up

Net force points with velocity; the speed increases.

02 · Slow down

Net force opposes velocity; the speed decreases.

03 · Turn

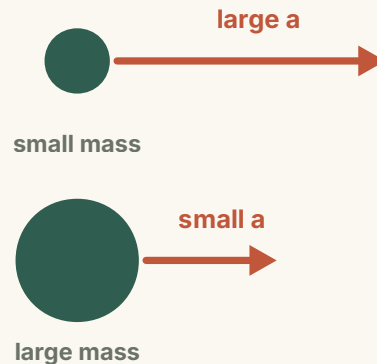
A sideways net force changes direction, even at constant speed.

DEFINE

Inertia is resistance to a change in velocity.

Every object has inertia. Mass measures how strongly it resists acceleration.

- 1 Inertia is a property**
Not a force — it does not need to be “activated”.
- 2 More mass, more inertia**
The same net force produces less acceleration.
- 3 Motion does not add inertia**
Same mass, same inertia — moving or at rest.



OBSERVE

Quickly remove the support; the object barely moves.

A short interaction gives little time for friction to change the horizontal velocity.

CARD AND COIN

Flick the card sideways; gravity then pulls the nearly stationary coin straight down into the glass.

TABLECLOTH AND DISHES

A fast pull limits the friction impulse, so the dishes stay nearly in place.

APPLY

A falling drop keeps the train's forward velocity.

Inside a constant-velocity train, the drop lands directly below its release point.

1 Before release

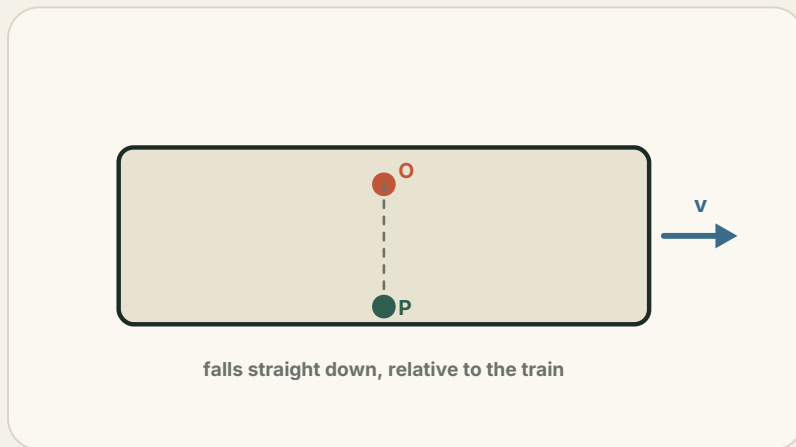
Drop and train share the same forward velocity.

2 During the fall

Gravity changes the vertical velocity, not the horizontal.

3 Result

Relative to the train, the drop falls straight from O to P.



DIAGNOSE

Loose objects reveal a vehicle's acceleration.

They tend to keep their previous velocity while the vehicle changes beneath them.

VEHICLE ACCELERATES FORWARD

Loose objects lag backwards; passengers lean back, liquid rises at the rear.

VEHICLE BRAKES

Loose objects continue forwards; passengers pitch forward, liquid rises at the front.

COMMON TRAPS

Repair three misleading statements.

Precise language prevents incorrect force diagrams and predictions.

01 · "Force keeps it moving."

Net force changes velocity; zero net force preserves it.

02 · "It moves because of inertia."

Inertia is a property, not a force — resistance to velocity change.

03 · "No net force = no forces."

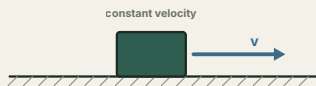
Several forces may act and balance to a zero vector sum.

● SUMMARY

Motion needs **no force to continue** — only to **change**.

ZERO NET FORCE $\Leftrightarrow$ CONSTANT VELOCITY

$$\sum \vec{F} = 0 \Leftrightarrow \vec{v} = \text{constant}$$



Everyday objects stop because resistance supplies that force.

Problem route: choose the object $\rightarrow$ list external forces $\rightarrow$ find the net force $\rightarrow$ decide if velocity changes.